

Information Collapse Theorem under Contradictory Inputs (Final)

From Semantic Opposition to Entropic Lower Bounds via Pairwise Boundary Distances

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1 Information Collapse Theorem under Contradictory Inputs (Final)

1.1 1. Introduction

When a classification model (e.g., a Transformer) processes natural language inputs containing semantic contradictions, algebraic cancellation may occur in the representation space,

preventing the output probability from concentrating on a single class and forcing unavoidable entropy growth. This theorem provides a conditional mathematical description of this phenomenon, built on three mutually dependent propositions.

The theorem is **conditional**: it holds only when its premises are satisfied. If the premises do not hold, the theorem does not trigger and its conclusions do not apply.

1.2 2. Preliminary Propositions

1.2.1 Proposition 1 (Algebraic Cancellation)

Let the representation space contain a pair of semantically opposed vectors $\vec{v}_A, \vec{v}_{-A}$ satisfying

$$\langle \vec{v}_A, \vec{v}_{-A} \rangle < 0.$$

Then the residual addition $\vec{v}_A + \vec{v}_{-A}$ induces norm compression, i.e.

$$\rho = \frac{\|\vec{v}_A + \vec{v}_{-A}\|}{\|\vec{v}_A\| + \|\vec{v}_{-A}\|} < 1.$$

This compression leads to dissipation of signal energy.

1.2.2 Proposition 2 (Topological Leakage via Pairwise Boundaries)

For a fixed sample representation z , let $i^* = \arg \max_i f_i(z)$ be the class with the largest logit.

For each competing class $j \neq i^*$, define the pairwise decision boundary

$$\mathcal{B}_{ij} = \{z' \in \mathbb{R}^d : f_i(z') = f_j(z')\},$$

and the distance from z to that boundary as

$$d_{ij}(z) = \inf_{z' \in \mathcal{B}_{ij}} \|z - z'\|.$$

Assume **strong geometric regularity**: for all $j \neq i^*$,

$$0 \in \mathcal{B}_{i^*j}, \quad \text{i.e.} \quad f_{i^*}(0) = f_j(0).$$

Then, because 0 is a feasible point for every relevant boundary, we have

$$d_{i^*j}(z) \leq \|z - 0\| = \|z\|, \quad \forall j \neq i^*.$$

Combining with Proposition 1, and letting $S = \|\vec{v}_A\| + \|\vec{v}_{\neg A}\| \leq S_{\max}$, we obtain

$$\max_{j \neq i^*} d_{i^*j}(z) \leq \|z\| = \rho S \leq \rho S_{\max}.$$

Define

$$\Phi(\rho) = \rho S_{\max}.$$

Then

$$\max_{j \neq i^*} d_{i^*j}(z) \leq \Phi(\rho), \quad \text{and} \quad \Phi(\rho) \rightarrow 0 \text{ as } \rho \rightarrow 0.$$

Thus algebraic cancellation pushes the representation into the transition zone of all pairwise decision boundaries, causing boundary leakage. This formulation avoids the need for global boundary geometry analysis.

1.2.2.1 Remark on Regularity Strengths

Condition	Geometric Property	Engineering Example	Probability Limit	Phenomenon
Weak regularity	Origin lies on some (but not all) pairwise boundaries	Non-zero bias; non-debiased LLM output layer	$p_{\max} \leq 1/2$, biased toward top logit at origin	Deterministic hallucina- tion
Strong regularity (required)	Origin lies on all pairwise boundaries for the top class	Zero-bias head; symmetrically regularized model	$p_{\max} \rightarrow 1/k$	Random hallucina- tion

The theorem relies on **strong regularity** for its main proof. Weak regularity gives a local limit result described after Proposition 3.

1.2.3 Proposition 3 (Probability Diffusion)

Let the logit function $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ be L -Lipschitz continuous. Assume strong regularity holds. Then for any competing class $j \neq i^*$,

$$f_{i^*}(z) - f_j(z) \leq 2L d_{i^*j}(z) \leq 2L\|z\| \leq 2L\Phi(\rho).$$

The Softmax maximum probability is

$$p_{\max} = \frac{1}{1 + \sum_{j \neq i^*} e^{-(f_{i^*}(z) - f_j(z))}}.$$

Using the bound above and the monotonicity of the exponential,

$$e^{-(f_{i^*}(z) - f_j(z))} \geq e^{-2L\Phi(\rho)}, \quad \forall j \neq i^*.$$

Therefore

$$p_{\max} \leq \frac{1}{1 + (k-1)e^{-2L\Phi(\rho)}} =: q(\rho).$$

This inequality maps energy dissipation in representation space into mass dispersion in probability space, flattening the output distribution.

1.2.3.1 Corollary (Conservative Bound with Active Competitors)

If only $m \leq k-1$ classes are truly competitive (i.e., their logit differences are within the bound), then

$$p_{\max} \leq \frac{1}{1 + me^{-2L\Phi(\rho)}}.$$

When $m < k-1$, this bound is numerically larger, hence more **conservative** than the bound using $k-1$, but potentially more realistic.

1.2.3.2 Extension: Weak Regularity and Local Probability Limit

If only weak regularity holds, the origin lies on some but not all pairwise boundaries $\mathcal{B}_{i^*j}$. Let $J_0 \subseteq \{j \neq i^*\}$ be the non-empty set such that

$$f_{i^*}(0) = f_j(0), \quad j \in J_0,$$

and for $j \notin J_0$, define

$$\Delta_j = f_{i^*}(0) - f_j(0).$$

Since i^* is the class with the largest logit at the origin, we have

$$\Delta_j \geq 0, \quad \forall j \notin J_0,$$

with strict positivity if $f_j(0) < f_{i^*}(0)$.

As $\|z\| \rightarrow 0$ (i.e., $\rho \rightarrow 0$), the logits converge to $f(0)$, and the Softmax output converges to the distribution determined by $f(0)$. Let

$$Z = 1 + |J_0| + \sum_{j \notin J_0} e^{-\Delta_j}.$$

The exact limiting probabilities are

$$p_{i^*}(z) \rightarrow \frac{1}{Z} =: q,$$

$$p_j(z) \rightarrow q, \quad j \in J_0,$$

$$p_j(z) \rightarrow qe^{-\Delta_j}, \quad j \notin J_0.$$

Because $J_0 \neq \emptyset$, we have $|J_0| \geq 1$, hence

$$q = \frac{1}{1 + |J_0| + \sum_{j \notin J_0} e^{-\Delta_j}} \leq \frac{1}{1 + |J_0|} \leq \frac{1}{2}.$$

Thus under weak regularity, in the local limit around the origin the maximum probability is strictly bounded below $1/2$, and the model cannot concentrate probability mass on any single class.

The corresponding limiting output entropy is

$$H_{\text{lim}} = \log Z + q \sum_{j \notin J_0} e^{-\Delta_j} \Delta_j.$$

Since $\Delta_j \geq 0$, the second term is non-negative, and therefore

$$H_{\text{lim}} \geq \log Z \geq \log(1 + |J_0|) \geq \log 2.$$

Note: This conclusion describes the **local limit behavior** as the input representation tends to the origin. It is **not** a global conditional entropy lower bound over the entire input distribution. To obtain a global lower bound, the strong regularity condition of the main theorem is required, or additional assumptions about the concentration of input mass near the origin must be imposed.

1.3 3. Theorem Statement

Theorem (Information Collapse Lower Bound under Contradictory Inputs)

Let the classification model have output dimension $k \geq 2$, logit function f be L -Lipschitz, and satisfy strong geometric regularity (all pairwise decision boundaries relevant to the top class pass through the origin). Let the sum of the norms of the cancelling vectors be bounded by $S_{\max}$, i.e. $\|\vec{v}_A\| + \|\vec{v}_{-A}\| \leq S_{\max}$. If the input X activates at least one pair of semantically opposed vectors $\vec{v}_A, \vec{v}_{-A}$ with negative inner product, and Propositions 1–3 hold, then the conditional entropy of the output satisfies

$$H(Y|X) \geq h\left(\frac{1}{1 + (k-1)e^{-2L\rho S_{\max}}}\right) > 0,$$

where $h(p) = -p \log p - (1-p) \log(1-p)$, and ρ is the norm compression ratio from Proposition 1.

Using the conservative bound with m active competitors, we have

$$H(Y|X) \geq h\left(\frac{1}{1 + me^{-2L\rho S_{\max}}}\right) > 0.$$

Justification of the entropy lower bound: Given $p_{\max} \leq q$, the minimal possible entropy occurs when the remaining probability mass $1-q$ is concentrated on a single other class. This

yields exactly $h(q)$. Therefore $h(q)$ is a **universal, tight lower bound** that does not require any symmetry assumption. The expression $h(q) + (1 - q) \log(k - 1)$, which would correspond to uniform distribution of the remaining mass, is actually an **upper bound** and cannot be used as a guaranteed lower bound.

Equivalently, the entropy loss

$$\Delta = \log k - H(Y|X) > 0$$

has a positive lower bound. If subsequent layers preserve bounded Lipschitz continuity, the lower bound propagates to the final output; otherwise, the data processing inequality ensures that mutual information does not increase, but entropy lower bound propagation requires additional assumptions.

1.4 4. Key Corollaries

1.4.1 4.1 Routine vs. Complex Tasks

- **Routine tasks:** No significant opposing vectors; premises do not trigger. Representations stay far from boundaries, logits separate well, Softmax concentrates, model stable.
- **Complex tasks:** Multiple conflicting constraints activate opposition, trigger cancellation. Representations pushed into transition zone, entropy lower bound locks in, model cannot determine a unique answer $\rightarrow$ hallucination.

1.4.2 4.2 Architectural Inevitability of Hallucination

When contradiction is strong ($\rho \rightarrow 0$), $p_{\max}$ is driven towards its limit. Depending on $f(0)$, two hallucination modes appear:

Regularity	Probability Limit	Entropy Lower Bound	Phenomenon
Weak (origin on some boundaries)	$p_{\max} \leq 1/2$, biased	local limit $\log 2$	Deterministic, repetitive
Strong (origin on all relevant boundaries)	$p_{\max} = 1/k$, uniform	global $h(1/k)$	Random gibberish

In both cases, extra input perturbations cannot shift the centroid; sampling behaves like isotropic or biased noise.

1.4.3 4.3 Ineffectiveness of Prompt Engineering

Adding more prompts does not reduce the lower bound; it may worsen cancellation. The only remedies are:

- Remove semantic contradictions (avoid triggering the theorem).
- Change architecture to break Lipschitz continuity or boundedness assumptions.

1.4.4 4.4 Positive Residual in Training Loss

The theorem implies $H_{\text{model}}(Y|X) > 0$ under contradiction. For the training loss, we distinguish two cases:

- **Case A (inherent label uncertainty)**

If the true distribution $P_{\text{true}}(Y|X)$ is non-degenerate, then $H_{\text{true}}(Y|X) > 0$. Since

$$\mathcal{L}_{\text{CE}} = H_{\text{true}} + \mathbb{E}_X[D_{\text{KL}}(P_{\text{true}}\|P_{\text{model}})] \geq H_{\text{true}} > 0,$$

the loss has a strictly positive lower bound independent of KL.

- **Case B (deterministic true label)**

If P_{true} is one-hot, then $H_{\text{true}} = 0$. However, the theorem forces P_{model} to be non-degenerate (its entropy is strictly positive). Hence $P_{\text{model}} \neq P_{\text{true}}$, and therefore

$$D_{\text{KL}}(P_{\text{true}} \| P_{\text{model}}) > 0.$$

Consequently,

$$\mathcal{L}_{\text{CE}} = 0 + \mathbb{E}_X[D_{\text{KL}}(P_{\text{true}} \| P_{\text{model}})] > 0.$$

In both cases, the cross-entropy loss has an incompressible positive residual. Gradient descent cannot eliminate this residual; it can only oscillate above the positive lower bound. This explains the validation loss plateau in complex reasoning tasks: it is not overfitting, but an architectural impossibility for the loss to decrease further.

1.5 5. Logical Status and Common Misunderstandings

1.5.1 5.1 Conditional Theorem

The theorem is of the form $(P_1 \wedge P_2 \wedge P_3) \Rightarrow Q$. Refutation must construct a case where all premises hold but Q fails. Pointing out that premises can be avoided is not a refutation.

1.5.2 5.2 Inseparability of Propositions

The three propositions form a pipeline: $P1 \rightarrow \rho$, $P2 \rightarrow d \leq \Phi(\rho)$, $P3 \rightarrow p_{\text{max}} \leq q$. A counterexample must break the entire chain.

1.5.3 5.3 Terminology

The “transition zone” is defined geometrically as $\{z : \|z\| < \delta\}$, not as “probability near uniform.” Probability diffusion is a consequence, not a definition. Mathematics only requires self-consistency.

1.6 6. Conclusion

This theorem provides a conditional guarantee: when semantic contradictions trigger algebraic cancellation and push the representation into the boundary transition zone, any finite Lipschitz classifier must produce a positive entropy lower bound. This offers a mathematical foundation for understanding hallucinations in large models, reveals the fundamental limitation of prompt engineering under contradictions, and shows that the positive residual in training loss is an architectural bottleneck, not just an inference issue.

Extended conclusion: Under weak regularity, the local output entropy limit as the representation approaches the origin is at least $\log 2$, exhibiting biased probability diffusion (deterministic hallucination). Strong regularity leads to uniform diffusion and random hallucination. The weak regularity result is a local limit and does not constitute a global conditional entropy lower bound; the global lower bound is provided by the main theorem under strong regularity.